



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	MECHANICS -I				
Type of Course	Core in Major				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY1CJ101: Fundamentals of Physics				
Course Summary	This course explores Newton's Laws of Motion and how they can be applied to solve different mechanical systems.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the concepts of linear and angular momentum, and dynamics of linear and rotational motion	U	C	Instructor-created exams / Quiz
CO2	Understand the concepts of the conservation laws of linear and angular momentum	U	C	Instructor-created exams / Quiz
CO3	Analyse collisions of particles using the conservation of linear momentum	An	P	Instructor-created exams / Home Assignments

CO4	Analyse rotating systems using the conservation of angular momentum	An	P	Instructor-created exams / Home Assignments
CO5	Demonstrate critical thinking and problem-solving skills by applying the concepts and techniques learned to solve an extended set of real-world problems.	Ap	P	Seminar Presentation / Group Tutorial Work
CO6	Understand the concepts of gravitational field, potential, and the Newtonian field equations, and apply them to analyse the gravitational effects of point masses and extended bodies.	U/Ap	C/P	Instructor-created exams / Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	MOMENTUM, IMPULSE AND COLLISIONS		11	16
	1	Momentum and Impulse	2	
	2	Conservation of Momentum	2	
	3	Momentum Conservation and Collisions	2	
	4	Elastic Collisions	2	
	5	Centre of Mass	1	
	6	Rocket Propulsion	2	
	Sections 8.1 – 8.6 of chapter 8 of Book 1			
II	ROTATION OF RIGID BODIES		12	18
	7	Angular Velocity and Acceleration	2	

	8	Rotation with Constant Angular Acceleration	2	
	9	Relating Linear and Angular Kinematics	2	
	10	Energy in Rotational Motion	2	
	11	Parallel-Axis Theorem	1	
	12	Moment of Inertia Calculations	3	
	Sections 9.1 – 9.6 of chapter 9 of Book 1			
III	DYNAMICS OF ROTATIONAL MOTION		12	18
	13	Torque	1	
	14	Torque and Angular Acceleration for a Rigid Body	2	
	15	Rigid Body Rotation about a Moving Axis	3	
	16	Work and Power in Rotational Motion	1	
	17	Angular Momentum	2	
	18	Conservation of Angular Momentum	2	
	19	Gyroscopes and Precession	1	
	Sections 10.1 – 10.7 of chapter 10 of Book 1			
IV	THE GRAVITATIONAL FIELD		13	18
	20	Newton's Law of Universal Gravitation	2	
	21	The Gravitational Field and Field of an Extended Body	3	
	22	The Gravitational Potential	3	
	23	Field Lines and Equipotential Surfaces	1	
	24	The Newtonian Gravitational Field Equations	3	

	25	The Equations of Poisson and Laplace	1	
	Sections 9.1 – 9.7 of chapter 9 of Book 2			
V	OPEN-ENDED MODULE: COMPUTATIONAL PROJECTS		12	
	Manageable number of selected computational projects from the list given may be assigned and evaluated. Any other computational projects related to the content of the course may be chosen by the teacher.			
	Computational Projects 1.1 – 1.4, 2.1 – 2.6, 3.1 – 3.3, 5.1 – 5.2, 6.1 – 6.6, 7.1, 9.1 – 9.4			
	Sections from References: Computational Projects in chapters 1, 2, 3, 5, 6, 7, 9 of Book 2			

Books and References:

1. University Physics with Modern Physics (Edn.15) by Young & Freedman (Book 1)
2. Intermediate Dynamics (Edn.2) by Patrick Hamill (Book 2)
3. An Introduction to Mechanics by Daniel Kleppner and Robert J. Kolenkow
4. Mechanics by Keith R. Symon
5. Mechanics: Berkeley Physics Course, Volume 1 by Charles Kittel, Walter D. Knight and Malvin A. Ruderman
6. Mechanics: From Newton's Laws to Deterministic Chaos by Florian Scheck
7. NPTEL video lectures: <https://nptel.ac.in/courses/115106090>

Mapping of COs with PSOs and POs :

	PSO1	PSO 2	PSO 3	PS O4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	1	0	3	1	3	0	0	1	3	0	0
CO 2	3	1	1	0	3	1	3	0	0	1	3	0	0
CO 3	3	2	3	0	3	1	3	0	1	1	3	0	0
CO 4	3	2	3	0	3	1	3	0	1	1	3	0	0
CO 5	3	0	3	1	2	1	3	2	1	1	3	0	0
CO 6	3	3	1	2	2	2	3	0	1	2	3	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	